

Lesson 10: Variational Quantum Classification

Feature maps, ansatz, optimization, and a complete VQC pipeline

Dr. Syed Muzamil Basha

School of Computer Science and Engineering, REVA University

Learning Outcomes

- Explain why classical data must be encoded into a quantum circuit.
- Distinguish a feature map from a variational ansatz.
- Understand the VQC training pipeline.
- Interpret the recorded 75% accuracy.

What Is a Variational Quantum Classifier?

- A VQC is a supervised classifier built from a parameterized quantum circuit.
- Classical input is encoded by a feature map.
- Trainable circuit parameters form the ansatz.
- An optimizer updates parameters to reduce classification error.

The Data Pipeline

- `make_classification` creates a two-feature synthetic dataset.
- `StandardScaler` standardizes the features.
- 20% of the data are held out for testing.
- The quantum circuit receives the two numerical features through a feature map.

Feature Map vs Ansatz

Feature Map

- Encodes data.
- Uses ZZFeatureMap.
- Includes entanglement.

Ansatz

- Contains trainable parameters.
- Uses RealAmplitudes.
- Learns a decision rule.

Core VQC Code

```
X, y = make_classification(n_samples=100, n_features=2,
                          n_informative=2, n_redundant=0,
                          n_clusters_per_class=1, random_state=42)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)
feature_map = ZZFeatureMap(feature_dimension=2, reps=2, entanglement='linear')
ansatz = RealAmplitudes(num_qubits=2, reps=3, entanglement='linear')
sampler = StatevectorSampler()
optimizer = COBYLA(maxiter=100)
vqc = VQC(feature_map=feature_map, ansatz=ansatz,
          optimizer=optimizer, sampler=sampler,
          pass_manager=pass_manager)
vqc.fit(X_train, y_train)
y_pred = vqc.predict(X_test)
accuracy = np.mean(y_pred == y_test)
print(f"VQC Accuracy: {accuracy * 100:.2f}%")
```

- `StatevectorSampler()` supplies the quantum sampling primitive used by the notebook.

Why Optimization Is Hybrid

- Quantum circuit evaluations produce information for the classical optimizer.
- COBYLA changes the trainable parameters.
- The loop repeats until the optimizer stops or reaches its iteration limit.
- This is a hybrid quantum-classical workflow.

VQC Accuracy: 75.00%

- The VQC correctly classifies 75% of the recorded test examples.
- The score is a test-set metric for this synthetic dataset.
- The result does not by itself prove a quantum advantage; it is a demonstration of the VQC workflow.

Beginner Activity

- Change the number of circuit repetitions (reps).
- Change the optimizer iteration limit.
- Observe how model accuracy can change when the circuit or optimization budget changes.

Lesson Summary

- Feature maps encode classical data.
- Ansatz parameters are optimized.
- Training and testing remain standard ML concepts.
- The notebook records 75% test accuracy.

**VQC combines quantum feature encoding, a trainable circuit,
and classical optimization.**

Quick Check

- ① What is a feature map?
- ② What is an ansatz?
- ③ Why is COBYLA used?
- ④ Why is VQC called hybrid?